



Perianal Crohn disease

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INTRODUCTION

Crohn disease is a chronic inflammatory condition that can affect any part of the gastrointestinal tract, from the oral cavity to the anus. Perianal manifestations of Crohn disease include perianal fistula, perianal abscess, anal canal lesions (anal fissures and anal stricture). Symptoms can vary from rectal pain and perianal discharge to bleeding or difficulty with defecation.

This topic will discuss the clinical features, diagnosis, and management of perianal manifestations of Crohn disease. The clinical features, diagnosis, and management of luminal Crohn disease are discussed in detail separately:

- (See "[Clinical manifestations, diagnosis, and prognosis of Crohn disease in adults](#)".)
- (See "[Overview of the medical management of mild \(low risk\) Crohn disease in adults](#)".)
- (See "[Medical management of moderate to severe Crohn disease in adults](#)".)

The diagnosis and management of anal fistula and perianal abscess in patients who do not have Crohn disease are discussed separately:

- (See "[Anorectal fistula: Clinical manifestations and diagnosis](#)".)
- (See "[Operative management of anorectal fistulas](#)".)
- (See "[Perianal and perirectal abscess](#)".)
- (See "[The role of imaging tests in the evaluation of anal abscesses and fistulas](#)".)

NORMAL ANORECTAL ANATOMY

The following is an overview of the anatomy of the anal region, while a more detailed description of the anal canal, ischiorectal fossa, perirectal tissues and sphincteric muscles can be found elsewhere (see "[Operative management of anorectal fistulas](#)", section on '[Anorectal anatomy](#)'):

- The anal canal, which is surrounded by internal (involuntary) and external (voluntary) sphincter muscles, extends 2.5 to 3.5 cm to the anal verge (visible lower edge of sphincter) inferiorly. In the mid-point of the anal canal is the dentate line, which is the demarcation between columnar epithelium superiorly and squamous epithelium inferiorly ([figure 1](#)). Along the dentate line lie crypts, which have small glands at their base. The squamous epithelium between the dentate line and anal verge is called "anoderm," which is similar to normal skin but highly sensitive, and the squamous epithelium outside the anal verge is the perianal skin.

INITIAL EVALUATION OF SYMPTOMS

The cause of perianal symptoms in patients with Crohn disease is identified by inspecting the perianal skin and anus, performing digital rectal examination, and when needed, a more detailed examination under anesthesia (EUA) ([table 1](#)):

- **Assess presenting symptoms** – Patients may report local symptoms such as perianal discharge (that may be purulent or bloody), rectal pain, rectal bleeding, fecal urgency, difficulty with defecation, or tenesmus. Patients with perirectal abscess may also present with general symptoms such as fever, chills, or malaise. Symptoms may be related to more than one perianal condition (eg, perianal fistula associated with perianal abscess).
- **Inspect perianal skin and anus** – Visual inspection of the perianal skin and anus precedes digital examination. Perianal inspection may reveal visible abnormalities such as a cutaneous fistula opening ([picture 1A-B](#)), external hemorrhoids ([picture 2](#)), or anal fissure ([picture 3](#)). For some patients, visual inspection is the extent of the physical examination because rectal pain precludes palpation of the perianal skin and digital rectal examination.
- **Perform digital rectal examination** – Digital rectal examination (DRE) is preceded by palpation of the perianal skin if a superficial abscess is suspected. Findings of perianal erythema and a palpable, often fluctuant, mass or nodule is suggestive of a superficial perianal abscess.

DRE is performed using topical anesthetic lubricant (eg, 2 percent [lidocaine](#) jelly) if rectal

pain is a presenting symptom. The lubricated pad of the examiner's finger is then slowly introduced through the anus and inserted into the rectum with circumferential palpation for any abnormalities (eg, fluctuance, submucosal fullness, narrowing of anal canal). Tenderness on digital examination is associated with perianal abscess, anal fissure, or hemorrhoids. Some patients cannot tolerate digital rectal examination because of rectal pain due to the underlying condition (eg, anal fissure, thrombosed hemorrhoid). If digital rectal examination cannot be performed due to discomfort, this is an indication for an EUA, which is often followed by radiologic imaging (eg, pelvic magnetic resonance imaging) to rule out an abscess. (See ["Anal fissure: Clinical manifestations, diagnosis, prevention"](#), section on 'Physical examination' and 'Pretreatment evaluation' below.)

PERIANAL FISTULA

Epidemiology — Perianal fistulizing disease affects approximately 5 to 40 percent of patients during the course of their Crohn disease [1-5]. In a population-based study of 946 patients with Crohn disease, 165 patients (17 percent) had at least one perianal fistula during a nearly 30-year period [5].

Risk factors — Disease severity, duration, and the presence of distal colonic disease are associated with a higher prevalence of perianal fistula. Involvement of the perianal region is more common in patients with colonic disease compared with those with disease restricted to the small bowel [1,2,6]. In one study, patients with Crohn colitis were greater than three times more likely to develop perianal fistulas compared with those with ileitis [7].

Perianal fistula is the initial manifestation of Crohn disease in approximately 10 percent of patients, however, most patients will develop additional symptoms (eg, diarrhea, abdominal pain) within one year [2,8]. (See ["Clinical manifestations, diagnosis, and prognosis of Crohn disease in adults"](#).)

Pathogenesis — In patients with Crohn disease, fistulas are caused by a penetrating abscess and/or inflammation into an adjacent organ or the skin. Thus, a fistula may initially present as an abscess that, upon spontaneous draining, evolves into a fistulous tract. The fistulous openings most commonly involve the perianal skin, but can also extend to other sites (eg, groin, vagina, bladder).

In contrast, an anorectal fistula that develops in patients without Crohn disease most commonly originate from an infected anal crypt gland. (See ["Anorectal fistula: Clinical manifestations and diagnosis"](#), section on 'Etiologies and pathogenesis'.)

Clinical features

Clinical presentation — Perianal discharge from a cutaneous opening is a commonly reported symptom, and the acuity of the presentation often suggests whether the fistula is complicated by an abscess.

- **Acute presentation** – Patients with perianal fistula associated with an abscess may present acutely with rectal pain, perianal tenderness or swelling, and fever. Purulent discharge from one or more cutaneous opening(s) may be reported. (See '[Perianal abscess](#)' below.)
- **Non-acute presentation** – Patients with a fistula without abscess may complain of nonpurulent perianal discharge from external opening(s) and may report nonacute rectal discomfort. Fistulas involving the bladder may be associated with pneumaturia or fecaluria, and females with fistulas to the vagina may report malodorous vaginal discharge. (See "[Colovesical fistulas](#)", section on '[Clinical manifestations](#)'.)

On physical examination, the skin may be erythematous, and a cutaneous opening may be visible or palpated as induration just below the skin. The cutaneous opening may be single or multiple, or it may drain fluid that is bloody, feculent, or purulent ([picture 1A-B](#) and [picture 4](#)). In some cases, an internal fistula opening can be palpated on digital rectal examination.

Anatomy and classification — Ways of categorizing perianal fistulas include a simple versus complex classification or the Parks classification system ([figure 2](#)). The former is more commonly used in clinical practice whereas the latter is used to guide operative management [2,9].

Fistulas can be classified as either simple or complex as follows (see '[Management](#)' below):

- **Simple perianal fistula** – A simple fistula is described as a low fistula (confined to the anal canal), with a single external opening and without abscess or stricture. This includes superficial fistula and intersphincteric fistula (Parks type 1).
- **Complex perianal fistula** – A complex perianal fistula is described as a high fistula that passes through or above muscle layers (eg, suprasphincteric fistula, which includes the entire sphincter apparatus), with single or multiple external openings, with or without abscess. If the fistula penetrates an adjacent organ such as the bowel or bladder, it is also considered complex.

Most simple perianal fistulas respond well to medical management (ie, antibiotics plus therapy for underlying luminal Crohn disease [eg, anti-tumor necrosis factor [TNF] agent]). Complex fistulas typically require a multidisciplinary approach that includes both medical therapy and surgical intervention. (See '[Management](#)' below.)

The Parks classification of perianal fistulas categorizes them into five types based on their relationship to the external anal sphincter and their surgical anatomy (ie, superficial, intersphincteric, transsphincteric, suprasphincteric, extrasphincteric) ([figure 2](#) and [figure 3](#)) [10,11]. This classification system describes the anatomic path of the fistula and is most useful for planning operative treatment. (See "[Operative management of anorectal fistulas](#)".)

The Parks classification system is discussed in more detail separately (see "[Anorectal fistula: Clinical manifestations and diagnosis](#)", section on '[Classifications](#)').

Diagnosis — The diagnosis of fistula is based upon characteristic findings on history and physical examination: cutaneous opening usually with perianal drainage with or without rectal pain. (See '[Clinical features](#)' above.)

While imaging studies are not required for diagnosis of simple fistulas, they are useful for excluding an abscess, defining the fistula tract in relation to the sphincter complex, and planning treatment for both simple and complex fistulas. (See '[Pretreatment evaluation](#)' below and "[The role of imaging tests in the evaluation of anal abscesses and fistulas](#)".)

Pretreatment evaluation — Therapy for perianal fistulas is guided by further evaluation (imaging with pelvic computed tomography [CT], pelvic magnetic resonance imaging [MRI] or rectal endoscopic ultrasound [EUS]; rectosigmoid endoscopy; examination under anesthesia [EUA]) that helps define the fistula anatomy and exclude the presence of a perianal abscess. (See '[Anatomy and classification](#)' above and "[The role of imaging tests in the evaluation of anal abscesses and fistulas](#)".)

The sequence of pretreatment testing depends on the acuity of the clinical presentation:

- Patients who present acutely with clinical findings suggestive of abscess (eg, fever, persistent perianal pain, purulent drainage, a palpable, fluctuant mass) first require prompt referral to a colorectal surgeon for EUA with abscess drainage and/or seton placement if needed. We also typically obtain pelvic CT with intravenous contrast only if an abscess is suspected. (See '[Patients with abscess](#)' below.)
- Patients without symptoms of an abscess undergo the following non-urgent pretreatment evaluation to define the fistula tract, exclude occult abscess, and assess

the rectosigmoid colon for active mucosal Crohn disease (see ['Perianal abscess'](#) below):

- **MRI of the pelvis (or rectal endoscopic ultrasound)** – Imaging studies for evaluating perianal fistula (pelvic MRI and rectal EUS) show air or contrast material within the fistula and help define the fistula tract ([image 1](#) and [image 2](#) and [image 3](#) and [image 4](#)). We generally obtain pelvic MRI because it is noninvasive and preferred for patients who present with perianal pain. MRI findings in addition to clinical features may be useful for predicting response to anti-TNF therapy for perianal Crohn disease [12].

The diagnostic accuracy and use of imaging tests for evaluating perianal fistula are discussed separately. (See ["The role of imaging tests in the evaluation of anal abscesses and fistulas"](#).)

- **Rectosigmoid endoscopy** – Rectosigmoid endoscopy assesses the rectal mucosa for active Crohn disease (eg, mucosal erythema, ulceration) and may identify the internal opening of the fistula. (See ["Clinical manifestations, diagnosis, and prognosis of Crohn disease in adults"](#), section on 'Endoscopy'.)

While the external opening of a fistula is usually visualized on the skin, the internal fistula opening may be seen endoscopically. The internal, mucosal opening of a low perianal fistula may be viewed with anoscopy, but sigmoidoscopy may be required to see a more proximal fistula with the internal opening in the rectum.

- **Examination under anesthesia** – During examination under anesthesia, the surgeon drains any sepsis encountered and probes the fistula tract with the intent of placing a noncutting seton. EUA is especially important for patients who cannot tolerate digital rectal examination due to perianal pain. In addition, rectosigmoid endoscopy is usually performed at the time of EUA. (See ['Management'](#) below.)

Obtaining at least two studies (EUA and MRI pelvis or rectal EUS) as part of pretreatment evaluation improves accuracy of determining the anatomy of the fistula. In a study of 32 patients with perianal Crohn disease, visualizing the perirectal space with two different modalities (EUS, MRI, or examination under anesthesia) accurately defined the anatomy for all fistulas compared with a consensus opinion of the anatomy based on all diagnostic studies [13].

Management

Goals and sequence of therapy — The following principles guide our approach to management of patients with perianal fistulas:

- **Goals of therapy** – Complete fistula closure is the primary therapeutic goal for most patients. For some patients with complex perianal fistulas, the achievable goal of therapy is symptomatic improvement (eg, less rectal pain, reduced drainage) and better quality of life but without complete fistula healing and closure [14]. (See ['Patients with nonhealing fistula'](#) below.)
- **Sequence of therapies** – The general sequence of multidisciplinary interventions is to eradicate the infection, if present; assess the status of luminal Crohn disease and the fistula tract in order to initiate medical therapy; and intervene surgically if needed (eg, nonhealing fistula).
- **Preference for therapies** – There is no single preferred treatment strategy for patients with perianal fistula, and various medical and surgical options are available. Our preference for therapies depends on the severity of the patient's symptoms, the anatomic complexity of the fistulas, the risk of adverse events, activity of the underlying Crohn disease, and patient preferences. Much of the data on the treatment of fistulas in Crohn disease is based upon case series, with few controlled trials.

The degree of inflammation and distribution of mucosal Crohn disease factors into overall management. Active inflammation in the rectum impairs fistula healing, and we optimize medical therapy to heal mucosal disease and to achieve clinical, endoscopic, and histologic remission. The approach to treating luminal Crohn disease is discussed separately. (See ["Overview of the medical management of mild \(low risk\) Crohn disease in adults"](#) and ["Medical management of moderate to severe Crohn disease in adults"](#).)

- **Multidisciplinary approach** – Management of perianal fistulas, and complex fistulas in particular, usually requires input from multiple disciplines (gastroenterology, colorectal surgery, radiology).

Simple fistula

Patients without symptoms — Patients with simple fistulas who are asymptomatic may not require treatment. If a patient with a simple fistula does not complain of rectal pain or drainage, we generally do not begin specific treatment for the fistula. We continue to treat the patient's underlying Crohn disease. Many of these patients will remain asymptomatic or the fistula will spontaneously heal. In a study including 21 patients with Crohn disease and perianal fistulas, seven patients (33 percent) remained asymptomatic despite continued presence of the fistula, while eight patients (38

percent) had spontaneous healing during 10 years of follow-up [15].

Patients with symptoms

Initial therapy — For patients who are symptomatic (eg, rectal pain, perianal drainage), initial therapy consists of oral antibiotic therapy ([metronidazole](#) or [ciprofloxacin](#)) in conjunction with treatment of mucosal rectal inflammation due to underlying Crohn disease. For patients with simple perianal fistula, the need for anti-TNF agents is determined by the status of the underlying mucosal disease, which is discussed separately. (See '[Goals and sequence of therapy](#)' above and "[Overview of the medical management of mild \(low risk\) Crohn disease in adults](#)" and "[Medical management of moderate to severe Crohn disease in adults](#)".)

We begin [metronidazole](#) 500 mg twice daily and make adjustments based on clinical response and presence of adverse effects [16-19]:

- For patients who respond (ie, with cessation of drainage and closure of the fistula) after four weeks of [metronidazole](#), we decrease the dose to 250 mg three times per day, continue therapy for another four weeks, and then stop it.
- For patients who develop adverse effects with [metronidazole](#) 500 mg twice a day, we reduce the dose to 250 mg three times a day (or switch to [ciprofloxacin](#) 500 mg twice daily) for a four-week course of therapy [20]. Adverse effects that are commonly associated with metronidazole are gastrointestinal symptoms (eg, nausea, vomiting, diarrhea, anorexia), and toxicity of metronidazole is discussed separately. (See "[Metronidazole: An overview](#)", section on '[Toxicity](#)'.)
- For patients who do not respond to either [metronidazole](#) or [ciprofloxacin](#) after four weeks of therapy, options include initiating biologic therapy or surgical intervention. (See '[Drug therapy](#)' below.)

Antibiotics are associated with reduced fistula drainage, fistula closure, and symptomatic improvement for patients with perianal Crohn disease, although data supporting their efficacy are mainly limited to small trials and studies [17,18,21-23]. The selection and dosing of antibiotic therapy are based on available data, traditional practice patterns, and clinician preference. Fistula healing rates with [metronidazole](#) (1000 to 1500 mg daily) range from 0 to 56 percent, while fistula recurrence after discontinuing antibiotics is not uncommon (approximately 75 percent of patients in one small study) [17]. In a meta-analysis of three trials including 112 patients with perianal fistula, patients treated with [ciprofloxacin](#) were

more likely to have either reduced fistula drainage or closure compared with placebo (risk ratio 1.64, 95% CI 1.16-2.32) [21]. Ciprofloxacin (500 mg twice daily) was compared with metronidazole (500 mg twice daily) or placebo in a 10-week trial of 25 patients with perianal Crohn disease [20]. Rates of fistula closure were not significantly different for ciprofloxacin compared with metronidazole or placebo (30 versus 0 or 13 percent, respectively).

Recurrence after initial response — Some patients will have a fistula recurrence following initial improvement when antibiotics are discontinued [17]. Patients who initially responded to a course of antibiotics (ie, [metronidazole](#) or [ciprofloxacin](#)) are given a second course of antibiotic therapy for four weeks. (See '[Initial therapy](#)' above.)

Subsequent therapy — For patients with simple fistulas who do not respond to antibiotic therapy or who have fistula recurrence after initial healing, medical therapy with a biologic agent is initiated. (See '[Drug therapy](#)' below.)

Seton placement is also performed; however, fistulotomy is an alternative option for patients with a nonhealing, superficial fistula with minimal sphincter involvement, which rarely occurs. A fistulotomy involves laying open the fistula tract in its entirety, and healing rates after fistulotomy for patients with Crohn disease are high in most studies (80 to 100 percent) [9]. The fistulotomy procedure and its efficacy in patients without Crohn disease are discussed separately. (See "[Operative management of anorectal fistulas](#)", section on '[Fistulotomy for simple fistulas](#)'.)

Patients with simple perianal fistula who do not respond to oral antibiotics and/or EUA with draining seton placement are managed with the treatment approach used for patients with complex perianal fistula. (See '[Complex fistula](#)' below.)

Complex fistula

Patients with abscess — Patients who have perianal fistula complicated by abscess require prompt surgical drainage of the abscess that is performed during EUA with placement of a noncutting silastic seton (a drain that is threaded into the cutaneous orifice of a fistula, through the fistula tract, and into the rectum and anal canal) when needed, in addition to antibiotic therapy. We typically give a two-week course of oral [metronidazole](#) 500 mg twice daily (or [ciprofloxacin](#) 500 mg twice daily is an alternative) [24] ([algorithm 1](#)) (see '[Pretreatment evaluation](#)' above).

The clinical presentation and management of perianal abscess is discussed below. (See

['Perianal abscess'](#) below.)

After the abscess is drained and infection is treated, medical therapy with an anti-TNF agent can be initiated, and timing of seton removal is determined as outlined below. (See ['Patients without abscess'](#) below.)

Local sepsis in the pelvis leads to tissue destruction of the anal sphincter, and setons are placed for continuous drainage. The combination of a draining seton followed by therapy with anti-TNF-agent results in higher rates of fistula healing and longer duration of fistula closure compared with anti-TNF agents alone [25-28]. (See ['Drug therapy'](#) below.)

Patients without abscess

Examination under anesthesia — Patients with complex perianal fistula without abscess require a combined approach consisting of non-urgent EUA with seton placement and medical therapy (an anti-TNF-agent combined with antibiotic therapy) [25,26,29]. (See ['Pretreatment evaluation'](#) above and ['Drug therapy'](#) below.)

Timing for removal of setons relative to ongoing medical therapy is a joint decision involving the colorectal surgeon and gastroenterologist. After setons are removed, the degree of fistula closure is assessed clinically by reduced drainage. Imaging (eg, pelvic MRI, rectal endoscopic ultrasound) can also determine if the fistula tract remains open or inflamed [30]. Anti-TNF agents are continued as maintenance therapy for patients with Crohn disease and healed fistulas. (See ["Medical management of moderate to severe Crohn disease in adults", section on 'Maintenance therapy'](#).)

If the fistula does not heal despite these measures including dose-optimizing the anti-TNF agent, options include replacement of setons, further surgical intervention as discussed below, or switching to a different biologic agent (eg, [ustekinumab](#)). (See ['Patients with nonhealing fistula'](#) below.)

Dosing for anti-TNF agents and therapeutic drug monitoring are discussed separately. (See ["Treatment of Crohn disease in adults: Dosing and monitoring of tumor necrosis factor-alpha inhibitors"](#).)

Drug therapy — Initial drug therapy for complex perianal fistula without abscess consists of anti-TNF agent (eg, [infliximab](#), [adalimumab](#), [certolizumab pegol](#)) combined with antibiotic therapy. We typically use oral [metronidazole](#) 500 mg twice daily (or [ciprofloxacin](#) 500 mg twice daily is an alternative) for two weeks.

The approach to induction and maintenance therapy with anti-TNF agents with or without an immunomodulator for patients with Crohn disease is discussed in more detail separately. (See "[Medical management of moderate to severe Crohn disease in adults](#)", section on 'Induction therapy' and "[Medical management of moderate to severe Crohn disease in adults](#)", section on 'Maintenance therapy'.)

Dosing, monitoring and adverse effects of anti-TNF agents are discussed separately:

- (See "[Treatment of Crohn disease in adults: Dosing and monitoring of tumor necrosis factor-alpha inhibitors](#)".)
- (See "[Tumor necrosis factor-alpha inhibitors: An overview of adverse effects](#)".)
- (See "[Tumor necrosis factor-alpha inhibitors: Bacterial, viral, and fungal infections](#)".)
- (See "[Tumor necrosis factor-alpha inhibitors: Risk of malignancy](#)".)

Anti-TNF agents are effective for healing and closure of perianal fistula, although fistula status was not a primary endpoint for some of the randomized trials which evaluated the efficacy of anti-TNF agents for patients with Crohn disease [31-36]. In a trial of 94 patients with Crohn disease and perianal fistula, induction therapy with [infliximab](#) (5 mg/kg or 10 mg/kg) was associated with higher rates of complete fistula closure compared with placebo (55 or 38 percent, respectively, versus 13 percent) [31]. In a 54-week maintenance trial of 282 patients with fistulizing Crohn disease, patients given infliximab 5 mg/kg every eight weeks had higher rates of fistula closure compared with placebo (36 versus 19 percent) [32].

Combination of an anti-TNF agent with [ciprofloxacin](#) may offer more benefit compared with anti-TNF monotherapy [20,37,38]. In a trial of 76 patients with fistulizing Crohn disease, [adalimumab](#) combined with ciprofloxacin resulted in higher rates of clinical response (defined as a 50 percent reduction of fistula at 12 weeks) compared with adalimumab plus placebo (71 versus 47 percent) [38].

Maintenance therapy with [infliximab](#) appears to reduce the need for surgery or inpatient hospitalizations. In a trial of 195 patients with fistulizing Crohn disease who initially responded to infliximab therapy and were evaluated at week 54, infliximab (5 mg/kg every eight weeks) was associated with fewer inpatient hospitalization days (0.5 versus 2.5 days) and fewer surgeries and procedures (65 versus 126 procedures) compared with placebo [39].

Anti-TNF agents are preferred for long-term treatment of perianal fistula; however,

thiopurines ([azathioprine](#) [AZA] and [6-mercaptopurine](#) [6-MP]) may be beneficial when used either as an alternative therapy for patients who cannot tolerate or do not wish to start an anti-TNF agent, or as part of combination therapy with an anti-TNF agent [40]. A meta-analysis of five randomized controlled trials showed that AZA or 6-MP treatment resulted in higher rates of complete fistula healing or decreased discharge (54 versus 21 percent) [41]. (See "[Medical management of moderate to severe Crohn disease in adults](#)", section on 'Thiopurines' and "[Medical management of moderate to severe Crohn disease in adults](#)", section on 'Combination therapy'.)

There may be a role for other biologic agents for treating perianal Crohn disease, although studies are limited. Data from subgroup analyses from induction and maintenance trials suggested that [vedolizumab](#) and [ustekinumab](#) resulted in some benefit for patients with perianal fistula [42-48]. In a trial including 32 patients with Crohn disease and perianal fistula who were treated with standard vedolizumab dosing (ie, 0, 2, 6, 14 and 22 weeks), an additional vedolizumab dose at week 10 was not associated with higher rates of fistula healing after 30 weeks compared with standard dosing [49]. In a subgroup analysis of clinical trials comparing [upadacitinib](#), a JAK 1 inhibitor, with placebo in patients with Crohn disease and perianal fistula, upadacitinib resulted in some benefit at week 12 [50].

Patients with nonhealing fistula — For patients who have failed medical management (defined as persistent drainage with a visible external opening and an associated tract on imaging), surgical interventions beyond seton placement can result in long-term improvement [51,52]. The goals of surgery are to close the fistula while preserving fecal continence or to reduce symptoms by making management easier for the patient such as by transforming a complex fistula into one closer to the anus, which is controllable by a small pad.

Surgical planning includes the following:

- **Timing of intervention** – The timing of surgery depends on whether active inflammation in the distal rectum and anal canal is present. Patients with active proctitis are treated medically before surgical intervention for persistently draining fistula, because surgical outcomes are better when luminal disease is in remission [53]. (See "[Medical management of moderate to severe Crohn disease in adults](#)", section on 'Induction therapy'.)
- **Choosing an intervention** – The choice of intervention depends on several factors such as the location, type, and duration of the fistulizing disease. Surgical options

include: removal of the seton with or without an advancement flap (rectal or cutaneous); simple fistulotomy (provided that the fistula is subcutaneous); ligation of the intersphincteric tract (ie, LIFT procedure); or proctectomy with permanent fecal diversion (particularly for patients with debilitating fecal incontinence) [51,54]. These surgical approaches are described separately. (See "[Operative management of anorectal fistulas](#)".)

Other options for nonhealing perianal fistula include local injection of mesenchymal stem cells. The use of mesenchymal stem cells for patients with Crohn disease and nonhealing perianal fistula is supported by society guidelines and by randomized trials showing that fistula healing rates ranged from 50 to 83 percent at 6 to 12 months after treatment [52,55-59]. As an example, in a trial of 212 patients with Crohn disease and nonhealing complex perianal fistula, an injection of adipose-derived stem cells resulted in higher rates of fistula closure compared with placebo after six months (50 versus 34 percent) [57].

Hyperbaric oxygen has been studied for treating nonhealing perianal fistula in Crohn disease, but its use in this setting requires further study [60].

Prognosis

Cancer risk — Carcinoma (squamous cell cancer of the anus or colorectal adenocarcinoma) can rarely develop within the perianal fistulous tract [61-65]. In a cohort study including 2382 patients with fistulizing perianal Crohn disease, 19 patients (0.8 percent) were diagnosed with cancer in a fistula tract, and the mean time from diagnosis of Crohn disease to diagnosis of cancer was six years [65].

Carcinoma should be suspected in patients with longstanding fistulizing Crohn disease (ie, >10 years) and any of the following [61]:

- Refractory or worsening symptoms related to chronic, nonhealing fistulas containing copious granulation tissue and induration.
- New onset symptoms in patients with perianal fistulas previously in remission. Such symptoms include pain, discharge, and swelling.

For these patients, we perform examination under general anesthesia with biopsy and curettage of the fistula tract. If cancer is found, staging includes pelvic magnetic resonance imaging and CT scan of the chest, abdomen and pelvis, and management is similar to that for patients with anal or colorectal malignancies. (See "[Overview of the management of rectal adenocarcinoma](#)" and "[Treatment protocols for anal cancer](#)".)

A consensus statement has been published by the Treatment Optimization and Classification of Perianal Crohn Disease (TOpClass) consortium, and our approach is generally consistent with this guidance [61].

Risk of recurrence — For most patients with perianal fistula and Crohn disease, a multidisciplinary approach with medical therapy with an anti-TNF agent combined with surgical intervention is successful in alleviating symptoms, reducing fistula drainage, and healing fistulas. Therapy with anti-TNF agents results in maintenance of remission for both luminal and fistulizing Crohn disease [32,34-36,66], however, the risk of fistula recurrence in the era of biologic therapy has not been well defined. In a trial including 306 patients with fistulizing Crohn disease, approximately 40 percent of patients on [infliximab](#) maintenance therapy developed loss of response during the follow-up period [32]. Data for loss of response with other anti-TNF agents are limited. (See "[Medical management of moderate to severe Crohn disease in adults](#)", section on 'Maintenance therapy'.)

PERIANAL ABSCESS

Perianal abscess should be suspected in patients who present with constant pain in the anal or rectal area, fever or purulent discharge. Purulent rectal discharge may be noted if the abscess has begun to drain spontaneously or if it is associated with a perianal fistula. A perianal abscess in a patient with Crohn disease is often related to obstruction of a perianal fistula tract but may also originate from an infected anal crypt gland ([figure 4](#)).

A superficial anorectal abscess can be diagnosed on physical examination with findings of perianal erythema and a palpable, often fluctuant mass ([picture 1A, 1C](#)). A deeper abscess can be diagnosed by feeling a tender, often fluctuant mass internally on digital rectal exam, or by imaging studies, such as pelvic magnetic resonance imaging or ultrasound of the pelvis. The classification and clinical features of perianal abscess are discussed separately. (See "[Perianal and perirectal abscess](#)", section on 'Classification' and "[Perianal and perirectal abscess](#)", section on 'Clinical manifestations' and "[Perianal and perirectal abscess](#)", section on 'Diagnosis'.)

Treatment of perianal abscess in patients with Crohn disease includes prompt surgical drainage, seton placement for continued drainage (if the tract is identified), and a short course (ie, two weeks) of antibiotics (usually [metronidazole](#) or [ciprofloxacin](#)) [20,67]. If the perianal abscess is associated with a perianal fistula, medical management (anti-tumor necrosis factor agent) for fistulizing Crohn disease is started after the local infection is

controlled and the abscess has been drained. (See '[Complex fistula](#)' above.)

The medical and surgical management of perianal abscess and descriptions of the surgical interventions are discussed in more detail, separately. (See "[Perianal and perirectal abscess](#)", section on '[Management](#)'.)

ANAL FISSURES

Clinical features and diagnosis — An anal fissure is an ulcer in the lining of the anal canal distal to the dentate line ([picture 3](#)). Anal fissures in patients with Crohn disease are secondary to ulceration from underlying mucosal inflammation, in contrast to sporadic anal fissures, which are associated with increased internal anal sphincter pressure [68]. (See "[Anal fissure: Clinical manifestations, diagnosis, prevention](#)", section on '[Pathogenesis](#)'.)

Anal fissures in Crohn disease may be asymptomatic or present with bleeding, deep ulceration ([picture 5](#)), or anal pain, which may be worse with defecation. Characteristics of anal fissures in patients with Crohn disease include location other than the posterior midline (eg, lateral or anterior) and multiple, recurrent, or nonhealing fissures. In addition, hypertrophic, edematous skin tags may be present and may show granulomas if biopsied ([picture 1A-B](#)) [69]. (See "[Clinical manifestations, diagnosis, and prognosis of Crohn disease in adults](#)", section on '[Endoscopy](#)'.)

The diagnosis of anal fissure is suspected based upon clinical presentation and is confirmed with physical examination by either directly visualizing a fissure or reproducing the patient's presenting complaints (ie, anal pain) by gentle digital palpation of the anal verge. Clinical features of anal fissure and performing physical examination to confirm the diagnosis are discussed in more detail separately. (See "[Anal fissure: Clinical manifestations, diagnosis, prevention](#)", section on '[Clinical manifestations](#)' and "[Anal fissure: Clinical manifestations, diagnosis, prevention](#)", section on '[Diagnosis](#)'.)

Treatment

- **Initial therapy** – Initial therapy for anal fissures in patients with Crohn disease is similar to that in the general population, and the goals of therapy are both fissure healing and improving symptoms. Medical management includes supportive measures (warm sitz baths, topic analgesics such as [lidocaine](#) jelly) and a topical vasodilator ([nifedipine](#) or [nitroglycerin](#)). The anoderm should be kept clean and dry, but without excessive wiping or use of astringent cleaners. Initial therapy for anal fissures is discussed in more detail

separately (see ["Anal fissure: Medical management"](#)).

- **Subsequent therapy** – Fissures that do not heal despite supportive measures and topical therapy with vasodilators and calcium channel blockers may require additional therapy such as chemical denervation of the internal anal sphincter with botulinum toxin. Injection of botulinum toxin is typically performed on both sides of the fissure in the intersphincteric groove. (See ["Anal fissure: Surgical management"](#).)

In some patients, a nonhealing ulcer in the posterior midline of the anal canal may appear as a typical anal fissure; however, the ulcer is often related to underlying active inflammation from Crohn disease and should be treated with medical therapy, rather than an intervention that will decrease anal sphincter tone and increase the risk of incontinence. Lateral internal sphincterotomy is not performed due to the risks of injuring the sphincter mechanism, fecal incontinence, and poor wound healing. In addition, surgery should be avoided in patients with active proctitis as it predisposes to poor healing [70].

OTHER PERIANAL DISEASES

Anal stricture — Chronic mucosal inflammation associated with anorectal Crohn disease may lead to fibrosis and subsequently to anal stricture [71]. Symptoms suggestive of anal stricture include fecal urgency or frequency, and tenesmus or difficulty with defecation, although some patients may not report any symptoms and the stricture is seen incidentally at the time of endoscopy [9]. The evaluation of an anal stricture includes endoscopy with biopsies to determine if the stricture is inflammatory, fibrotic, or neoplastic, and examination under anesthesia with biopsies and pelvic magnetic resonance imaging are subsequently performed for some patients (eg, those with nondiagnostic endoscopic findings) [53].

A symptomatic, short fibrotic anal stricture without surrounding mucosal inflammation is managed with endoscopic balloon dilation or with dilation during examination under anesthesia (EUA), while inflammatory strictures are treated with medical therapy for luminal Crohn disease (eg, anti-TNF agent which is discussed separately) with or without mechanical dilation. (See ["Medical management of moderate to severe Crohn disease in adults"](#), section on 'Induction therapy'.)

Patients with an anorectal stricture that cannot be traversed with the endoscope should be referred for surgical consultation for EUA, stricture dilation, and consideration of resection. (See ["Surveillance and management of dysplasia in patients with inflammatory bowel](#)

disease", section on 'Strictures'.)

Hemorrhoids — Patients with Crohn disease may develop hemorrhoids, which are not related to their underlying Crohn disease. Patients with hemorrhoids commonly present with rectal bleeding, especially with bowel movements, or a sensation of rectal fullness. Therapy for hemorrhoids is local and conservative (ie, dietary and lifestyle modification, topical analgesics, and topical glucocorticoids), and this is discussed separately. (See ["Home and office treatment of symptomatic hemorrhoids"](#).)

Patients who have persistent symptoms despite initial therapy are referred to a colorectal specialist for consideration of treatment. Hemorrhoidectomy should be avoided because of the possibility of poor wound-healing and of damaging the anal sphincter.

SOCIETY GUIDELINE LINKS

Links to society and government-sponsored guidelines from selected countries and regions around the world are provided separately. (See ["Society guideline links: Crohn disease in adults"](#) and ["Society guideline links: Anal abscess and anal fistula"](#).)

INFORMATION FOR PATIENTS

UpToDate offers two types of patient education materials, "The Basics" and "Beyond the Basics." The Basics patient education pieces are written in plain language, at the 5th to 6th grade reading level, and they answer the four or five key questions a patient might have about a given condition. These articles are best for patients who want a general overview and who prefer short, easy-to-read materials. Beyond the Basics patient education pieces are longer, more sophisticated, and more detailed. These articles are written at the 10th to 12th grade reading level and are best for patients who want in-depth information and are comfortable with some medical jargon.

Here are the patient education articles that are relevant to this topic. We encourage you to print or e-mail these topics to your patients. (You can also locate patient education articles on a variety of subjects by searching on "patient info" and the keyword(s) of interest.)

- Basics topics (see ["Patient education: Crohn disease in adults \(The Basics\)"](#))
 - Beyond the Basics topics (see ["Patient education: Crohn disease \(Beyond the Basics\)"](#) and ["Patient education: Anal fissure \(Beyond the Basics\)"](#))
-

SUMMARY AND RECOMMENDATIONS

- **Background** – Perianal manifestations of Crohn disease include perianal fistula, perianal abscess, and anal canal lesions (anal fissures and anal stricture). Symptoms can vary from rectal pain and perianal discharge to bleeding or difficulty with defecation. (See ['Introduction'](#) above.)
- **Initial evaluation** – For patients with Crohn disease, the cause of perianal symptoms is identified by inspecting the perianal skin and anus, performing digital rectal examination, and when needed, a more detailed examination under anesthesia ([table 1](#)). (See ['Initial evaluation of symptoms'](#) above.)
- **Perianal fistula** – For patients with Crohn disease, fistulas are caused by a penetrating abscess and/or inflammation into an adjacent organ or the skin. The fistulous openings most commonly involve the perianal skin, but can also extend to other sites (eg, groin, vagina, bladder). (See ['Pathogenesis'](#) above.)
- **Diagnosis** – For patients with Crohn disease, diagnosis of a perianal fistula is based upon characteristic findings on history and physical examination: a cutaneous opening usually with perianal drainage with or without rectal pain. While imaging studies are not required for diagnosis of simple fistulas, they are useful for excluding an abscess, defining the fistula tract, and planning treatment. (See ['Diagnosis'](#) above and ["The role of imaging tests in the evaluation of anal abscesses and fistulas"](#).)
- **Management** – For most patients with Crohn disease and perianal fistula, complete fistula closure is the primary therapeutic goal. For some patients with complex perianal fistulas, the achievable goal of therapy is symptomatic improvement (eg, less rectal pain, reduced drainage) and better quality of life but without complete fistula healing and closure. (See ['Goals and sequence of therapy'](#) above.)
 - Simple fistula – For patients with simple fistula who are asymptomatic, we generally do not begin specific treatment for the fistula, while we continue to treat underlying Crohn disease. (See ['Patients without symptoms'](#) above.)

For patients with Crohn disease and perianal fistula who are symptomatic (eg, rectal pain, perianal drainage), we suggest antibiotic therapy in addition to treatment for underlying Crohn disease rather than no antibiotic therapy (**Grade 2C**). For patients with simple fistula, we typically use oral [metronidazole](#) 500 mg twice daily for four weeks and make dosing adjustments based on clinical response and patient tolerance. (See ['Initial therapy'](#) above.)

- **Complex fistula** – For patients with complex perianal fistula, examination under anesthesia is first performed for abscess drainage and seton placement (if needed). We typically use oral [metronidazole](#) 500 mg twice daily (or [ciprofloxacin](#) 500 mg twice daily) for two weeks in addition to anti-tumor necrosis factor (TNF) therapy ([algorithm 1](#)). (See '[Complex fistula](#)' above and "[Medical management of moderate to severe Crohn disease in adults](#)", section on '[Induction therapy](#)'.)

For patients with perianal fistulizing Crohn disease who respond to induction therapy with an anti-TNF agent, we suggest continued maintenance with the anti-TNF agent (**Grade 2B**). (See '[Drug therapy](#)' above.)

- **Perianal abscess** – For patients with Crohn disease and perianal abscess, treatment includes prompt surgical drainage, seton placement for continued drainage (if the tract is identified), and a course of antibiotic therapy (typically [metronidazole](#) or [ciprofloxacin](#)). (See '[Perianal abscess](#)' above and "[Perianal and perirectal abscess](#)".)
- **Anal fissure** – For patients with Crohn disease and anal fissure, therapy includes supportive measures (warm sitz baths, topical analgesics such as [lidocaine](#) jelly) and a topical vasodilator ([nifedipine](#) or [nitroglycerin](#)). Surgery should be avoided in patients with active proctitis as it predisposes to poor healing. (See '[Treatment](#)' above and "[Anal fissure: Medical management](#)".)

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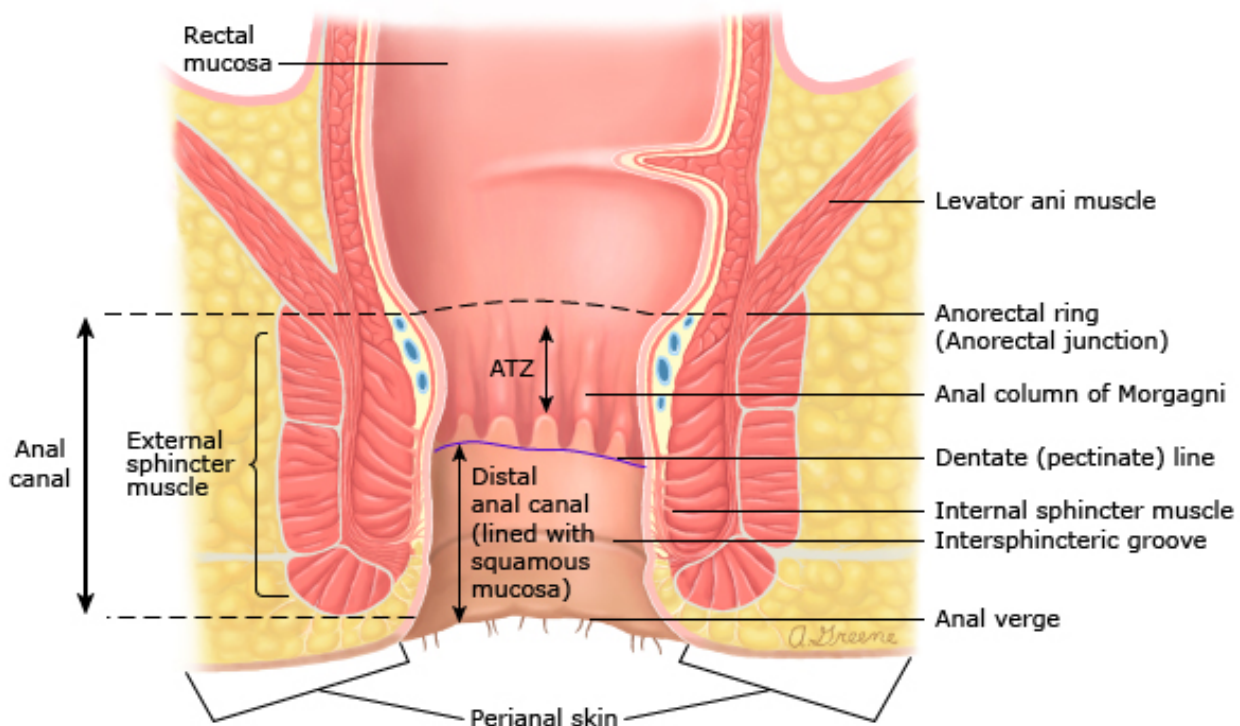
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Figure 1: Anatomy of the anus and rectum



The anal canal is 2.5 to 4.0 cm long and begins superiorly where the rectal ampulla is narrowed by the anorectal ring. This palpable muscular ring is formed by fusion of the puborectalis muscle (part of the levator ani muscle complex) with the more inferior internal and external anal canal sphincters.

The external anal canal sphincter ends just distally to the internal anal canal sphincter; the intersphincteric groove is the palpable plane that can be palpated between the termination of the two sphincters. The presence of the intersphincteric groove coincides roughly with the anal verge, which marks the distal portion of the anal canal. The perianal skin extends 5 cm laterally from the anal verge and is characterized by the presence of hair follicles and glands.

The interior of the anal canal can be divided into proximal and distal portions by an irregular line formed by the anal valves called the dentate (or pectinate) line (colored purple in the diagram). The portions of the anal canal proximal and distal to the dentate line have different origins of arterial supply, nerve innervation, and venous lymphatic drainage. The squamo-columnar junction (SCJ) lies within the proximal portion of the anal canal and marks the transition between rectal columnar epithelium to anal squamous epithelium. The exact position of the SCJ changes with time due to replacement of columnar epithelium with squamous epithelium in a process known as squamous metaplasia. The anal transformation zone (ATZ) is the zone where all aspects of squamous metaplasia are currently found and/or have occurred. The ATZ is marked by the SCJ proximally and extends distally to approximately the level of the dentate line.

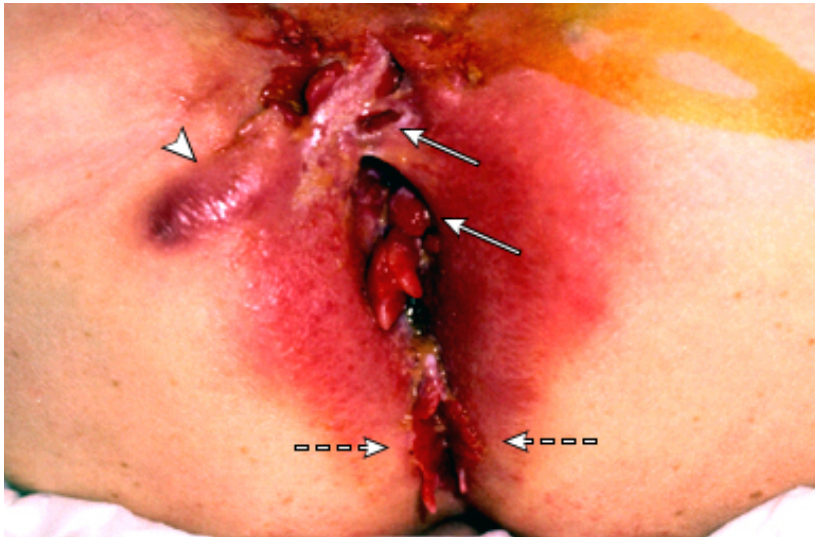
Table 1: Clinical features of perianal Crohn disease

Type of lesion	Typical location	Common clinical presentations	Perianal inspection	Digital rectal examination
Simple perianal fistula	Perianal skin	■ Perianal drainage	■ Single cutaneous opening	■ Internal opening may be palpated
Complex perianal fistula	Perianal skin or buttocks	■ Perianal drainage ■ Perianal pain	■ One or more cutaneous opening(s)	■ Internal opening(s) may be palpated
Perianal abscess	Varies*	■ Perianal pain, unrelated to bowel movements ■ Fluctuation beneath perianal skin ■ Cutaneous drainage of pus from fistula ■ Fever may be present	■ Cutaneous area of erythematous induration and/or fluctuance ■ Cutaneous opening if associated with fistula or spontaneous drainage	■ Diffuse tenderness may be present ■ Palpable fluctuant mass or fullness
Anal fissure	Anal canal	■ Rectal bleeding possible ■ May be painless	■ Tear or laceration in the anoderm ■ Bleeding may be visible	■ Tenderness of anal canal may be present
Anorectal stricture	Anal canal	■ Fecal urgency and/or frequency ■ Tenesmus ■ Difficult evacuation	■ Normal	■ Narrowing of anal canal

This table does not substitute for the clinical judgment of the treating specialist. Refer to UpToDate content on the diagnosis and management of perianal Crohn disease.

* The location of perianal abscess varies (eg, superficial, ischiorectal, intersphincteric).

Picture 1A: Anal fistulas and abscess in Crohn disease



This picture shows anterior perianal fistulas involving the vulva and anterior perineum (arrows) in a woman with Crohn disease being prepared for surgery. The surrounding skin is erythematous and indurated. An abscess (arrowhead) appears as a localized swelling. Hypertrophic skin tags (dashed arrows) in the anal canal are commonly observed in perianal Crohn disease and may be confused with external hemorrhoids.

Courtesy of Alain Bitton, MD, FRCPC.

Picture 1B: Perianal fistula

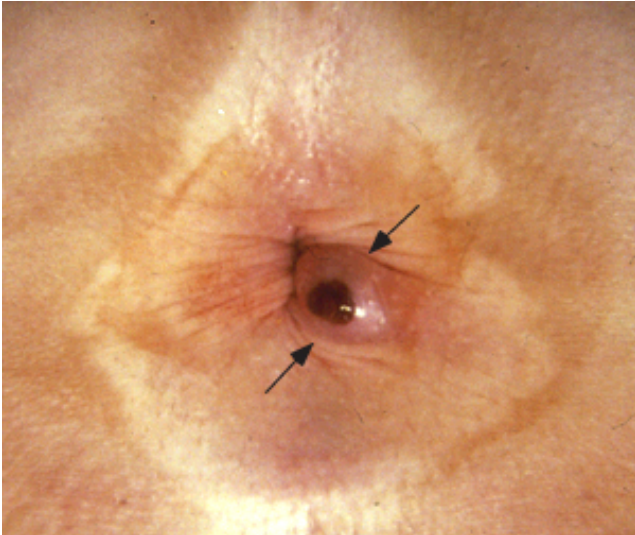


Perianal fistula in a patient with Crohn disease. Scarring and deformity from previous fistulas is also apparent.

Courtesy of David A Schwartz, MD and Maurits J Wiersema, MD.

Graphic 78460 Version 2.0

Picture 2: Thrombosed external hemorrhoids



Photograph shows a swollen external hemorrhoid (arrows).

Reproduced with permission of the American Society of Colon and Rectal Surgeons.

Graphic 72916 Version 2.0

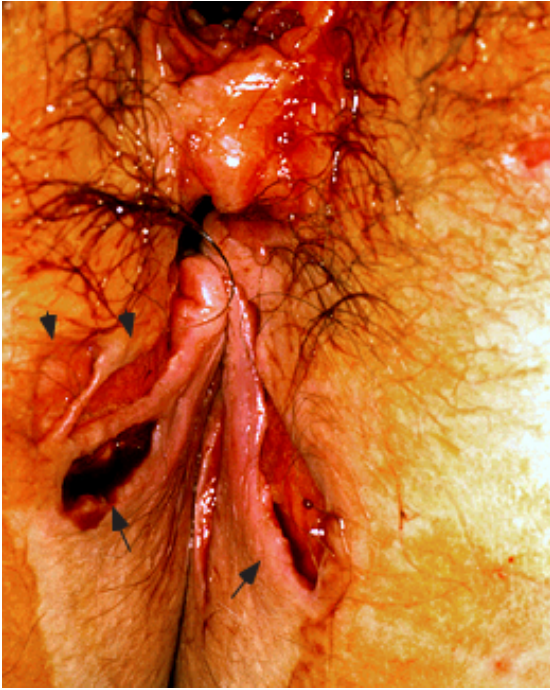
Picture 3: Chronic anal fissure



Photograph of a chronic anal fissure in the posterior midline (arrow), which is the most common site of fissure formation. The raised edges and fibrotic appearance at the base of this fissure distinguish it from an acute anal fissure, which appears like a fresh laceration.

Reproduced by permission from the American Society of Colon and Rectal Surgeons.

Picture 4: Perianal Crohn disease

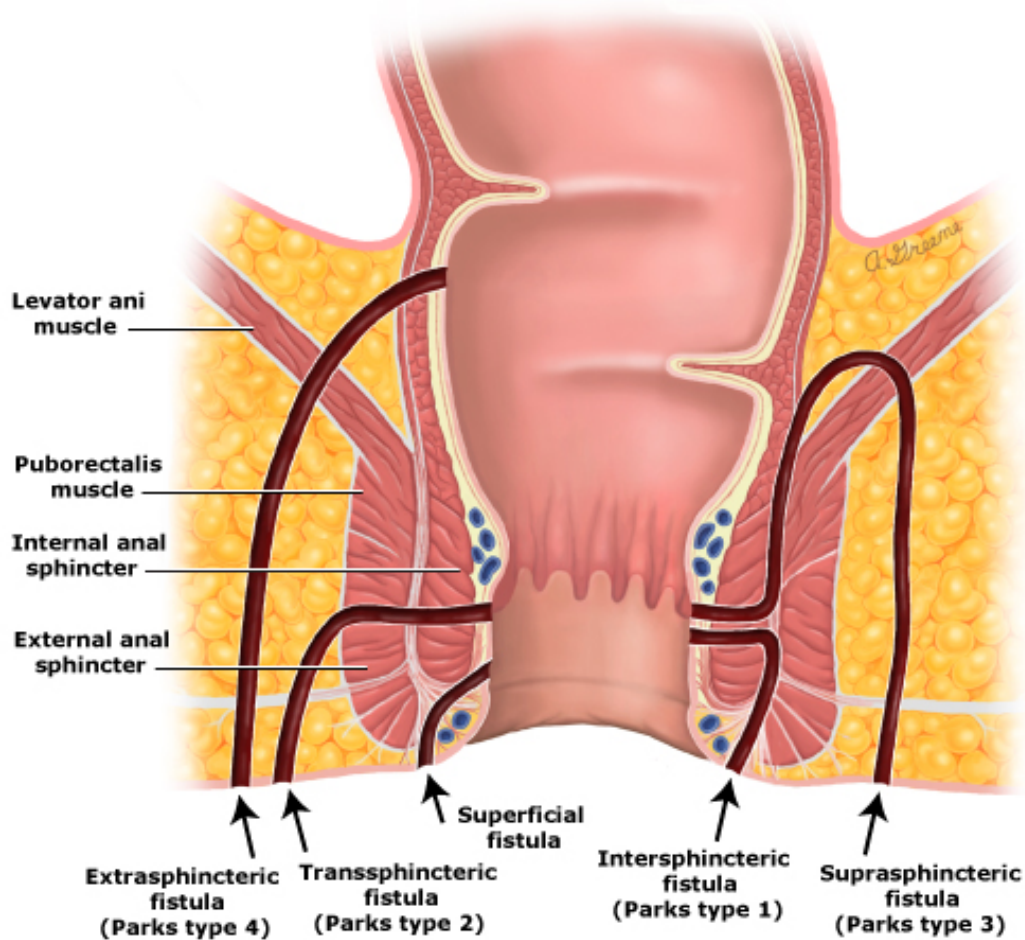


Perianal fistulas and ulcers in a patient with Crohn disease. There are bilateral posterior perianal fistulas (arrows) and two right posterior ulcers (arrowheads).

Courtesy of Alain Bitton, MD, FRCPC.

Graphic 65699 Version 2.0

Figure 2: Parks' classification of anorectal fistulas, anterior view



Parks described four types of anorectal fistulas that originate from cryptoglandular infections. Fistulas can have a complicated anatomy with one or more extensions and accessory tracts. The original Parks' classification did not include a superficial fistula tract.

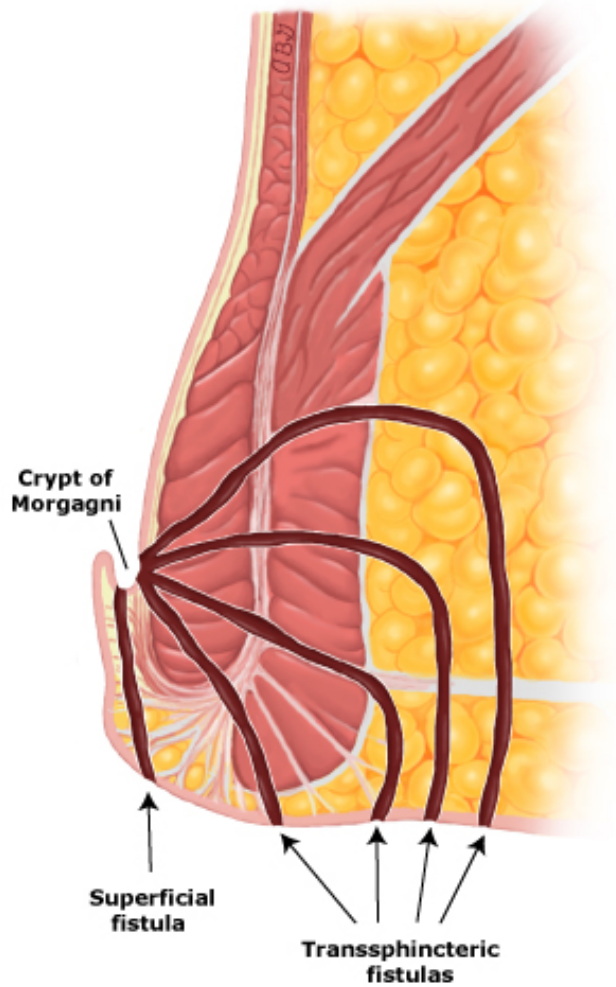
Type 1 is an intersphincteric fistula that travels along the intersphincteric plane.

Type 2 is a transsphincteric fistula that encompasses a portion of the internal and external sphincter.

Type 3 is a suprasphincteric fistula that encompasses the entire sphincter apparatus.

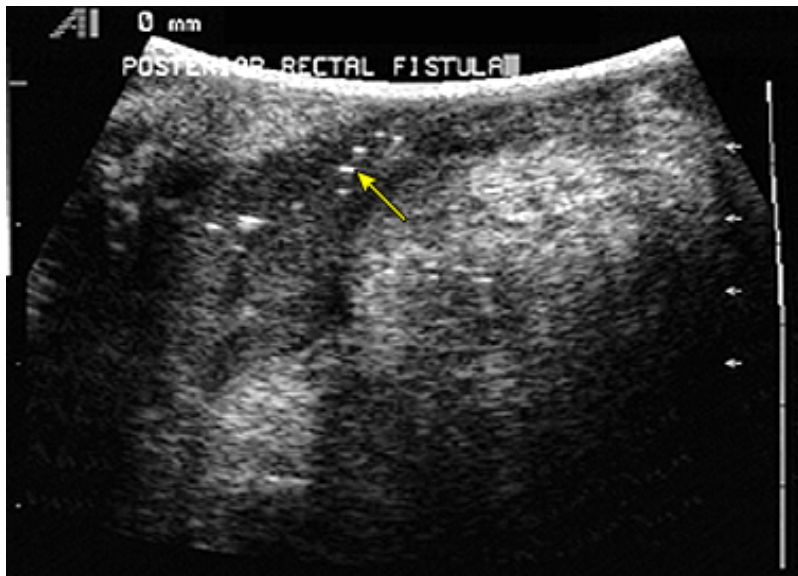
Type 4 is an extrasphincteric fistula that extends from a primary opening in the rectum, encompasses the entire sphincter apparatus, and opens onto the skin overlying the buttock.

Figure 3: Anorectal fistulas, lateral view



In this lateral view of anorectal fistulas, a superficial fistula has been included. It is not a component of the original Parks' classification. This figure also illustrates the potential pathways of transsphincteric fistulas.

Image 1: Air in fistula tract

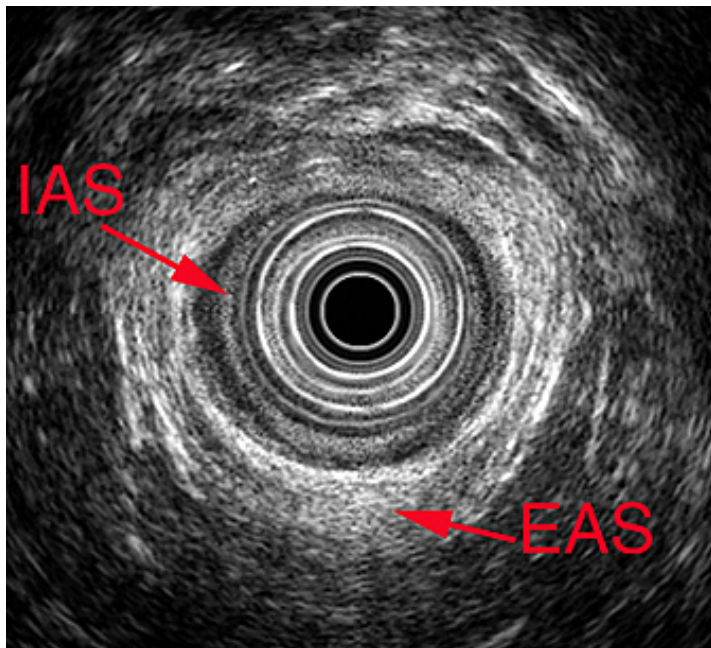


Endoscopic ultrasound image showing air in a fistula tract (arrow).

Courtesy of David A Schwartz, MD and Maurits J Wiersema, MD.

Graphic 56772 Version 3.0

Image 2: Endoscopic ultrasound of the anal sphincter

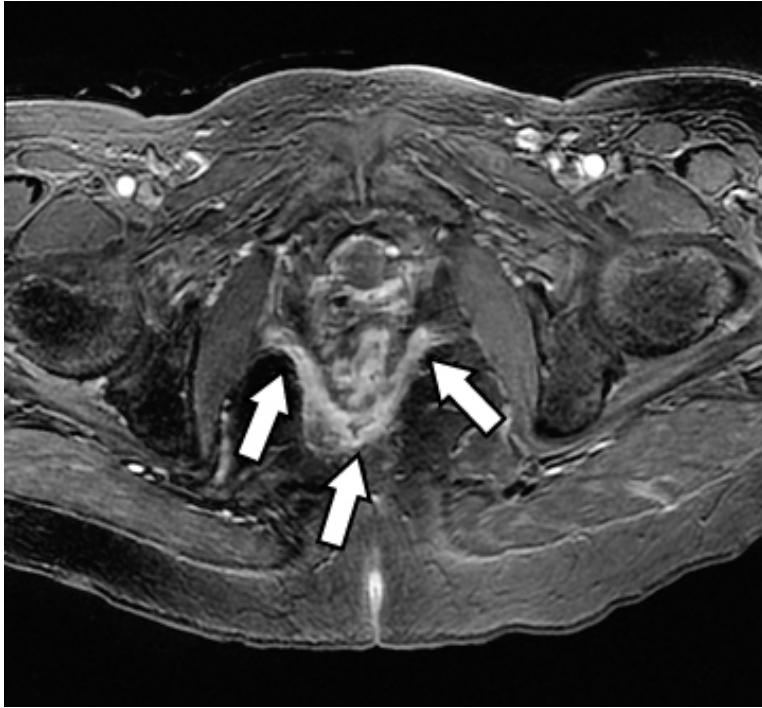


Endoscopic ultrasound image demonstrating the internal (IAS) and external (EAS) anal sphincters.

Courtesy of David A Schwartz, MD and Maurits J Wiersema, MD.

Graphic 56203 Version 3.0

Image 3: Perianal fistulas in Crohn disease

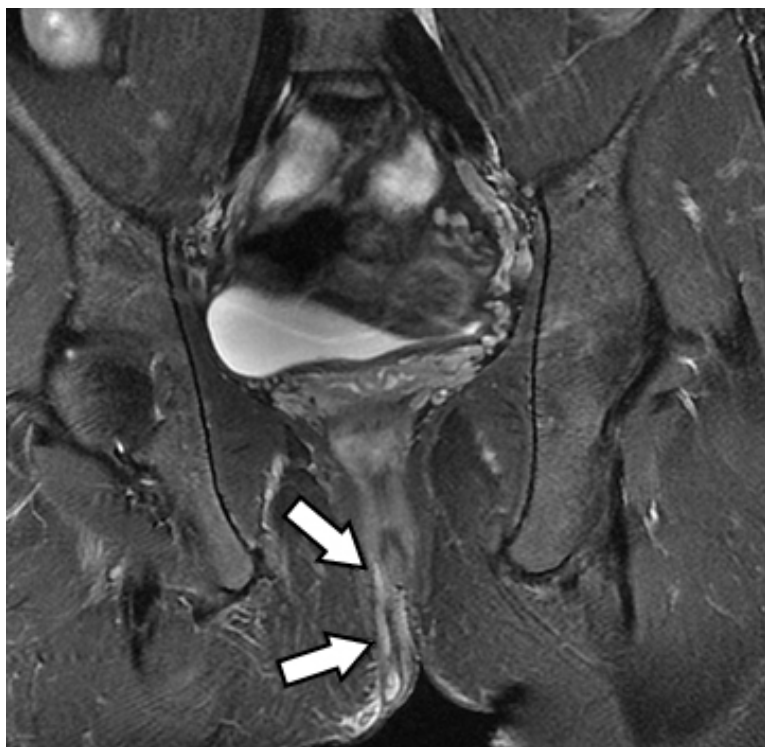


T2-weighted axial magnetic resonance imaging (MRI) study showing perianal fistulas (arrows) in a patient with Crohn disease.

Courtesy of Jonathan Kruskal, MD.

Graphic 72435 Version 5.0

Image 4: Perianal fistulas in Crohn disease

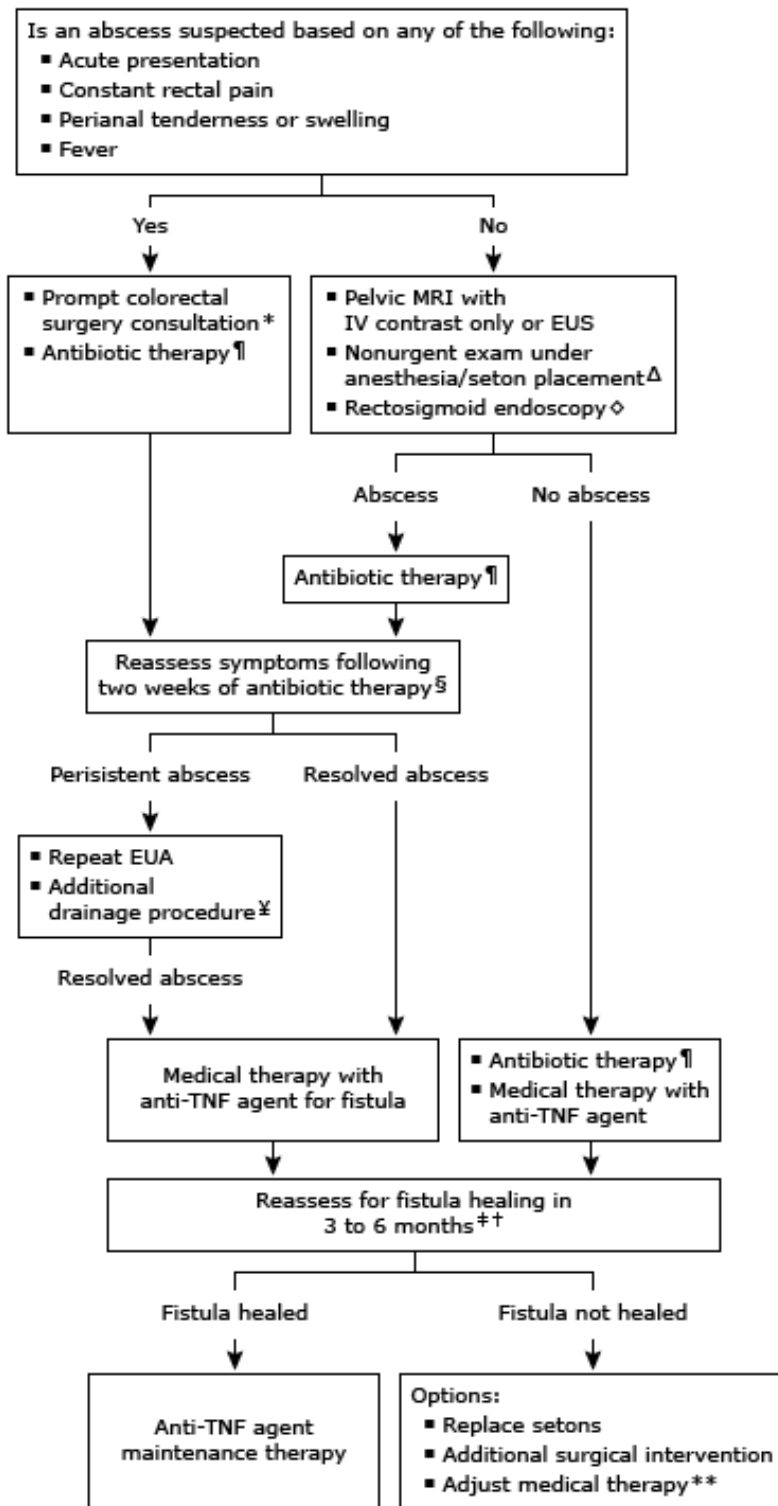


T2-weighted coronal magnetic resonance imaging (MRI) study showing a perianal fistula (arrows) in a patient with Crohn disease.

Courtesy of Jonathan Kruskal, MD.

Graphic 50476 Version 5.0

Algorithm 1: Management of complex perianal fistula in patients with Crohn disease



This algorithm on the management of complex perianal fistula in patients with Crohn disease does not substitute for the clinical judgment of the treating specialist. A complex perianal fistula is described as a high fistula that passes through or above muscle layers with single or multiple external openings, with or without abscess. Refer to UpToDate content on perianal Crohn disease.

MRI: magnetic resonance imaging; EUA: exam under anesthesia; EUS: endoscopic ultrasound; TNF: tumor necrosis factor; CT: computed tomography.

* Colorectal surgery is consulted for EUA with abscess drainage. Rectosigmoid endoscopy is often performed during EUA to assess mucosal disease. Pelvic CT scan with intravenous contrast only is typically performed if abscess is suspected.

¶ Antibiotic therapy typically consists of either oral metronidazole or ciprofloxacin for two weeks.
Δ The timing for removal of setons relative to ongoing medical therapy is a joint decision involving the colorectal surgeon and gastroenterologist.

◇ Rectosigmoid endoscopy is often performed at the time of EUA.

§ Symptomatic improvement (eg, absence of rectal pain, fever) suggests abscess resolution.

¥ For patients with persistent perianal abscess despite initial drainage, subsequent intervention may include surgical drainage or interventional radiology procedure. After abscess is drained and infection is treated, medical therapy with anti-TNF agent can be initiated.

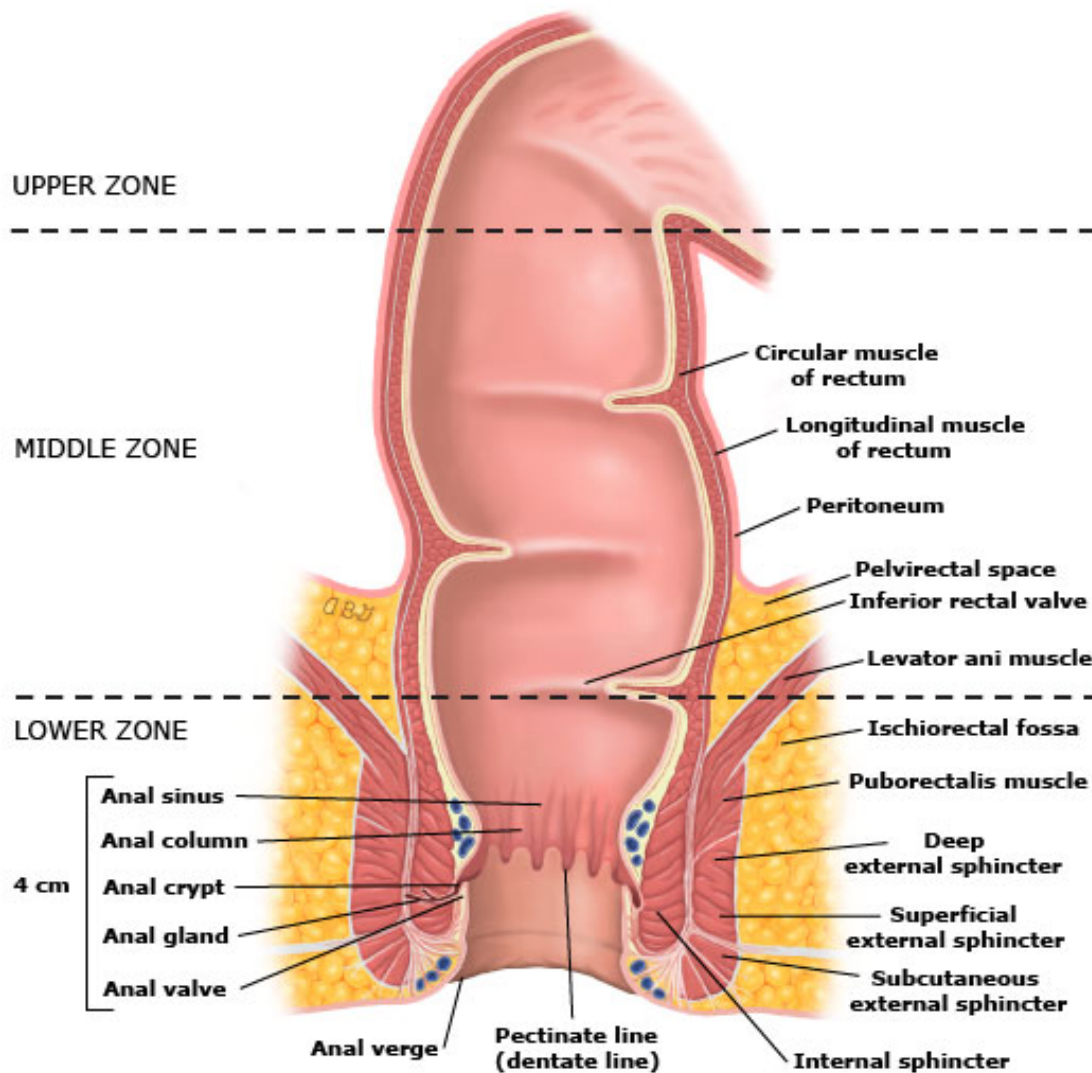
‡ Complete fistula closure is the therapeutic goal for most patients. For some patients with complex perianal fistulas, the achievable goal of therapy is symptomatic improvement (eg, less rectal pain, reduced drainage) and better quality of life but without complete fistula healing and closure.

† In addition to symptoms (eg, lack of drainage) and signs (eg, closure of external opening), options to assess fistula healing include pelvic MRI, rectal EUS and EUA. Testing method depends on initial presentation, prior testing, and clinician preference.

** Options for adjusting anti-TNF therapy for patients with nonhealing fistula include:

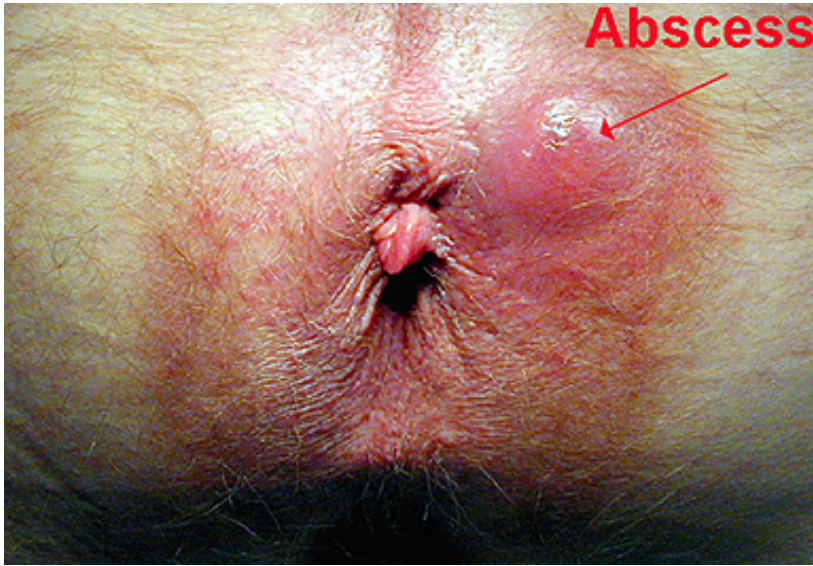
- Optimize dose of anti-TNF agent
- Begin a thiopurine (in addition to anti-TNF agent)
- Switch to a different anti-TNF agent
- Discontinue anti-TNF agent and begin an alternative biologic agent (eg, vedolizumab)

Figure 4: Anatomy of the anal canal and rectum



This figure illustrates the anatomy of the rectum and anal canal. Note the anal crypts and glands; 90% of anorectal fistulas originate in a cryptoglandular abscess. Also note the relationship of the crypts and glands to the internal and external sphincters.

Picture 1C: Perianal abscess



A perianal abscess is apparent as an erythematous, fluctuant bulge with surrounding edema.

Courtesy of David A Schwartz, MD and Maurits J Wiersema, MD.

Graphic 67524 Version 1.0

Picture 5: Perianal Crohn disease



Perineum in a woman with Crohn disease shows a large posterior ulcer (arrow) with anterior edematous fissure (arrowhead).

Courtesy of Alain Bitton, MD, FRCPC.

Graphic 80569 Version 4.0

Contributor Disclosures

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